



**DAYANANDA SAGAR
UNIVERSITY**



**SCHOOL OF
ENGINEERING**

**Bachelor of Technology
in
COMPUTER SCIENCE AND ENGINEERING**

23CS3503 – MACHINE LEARNING

MINI PROJECT REPORT

On

Traffic Flow Prediction

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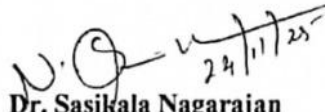


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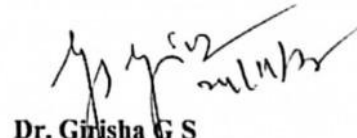
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CERTIFICATE

This is to certify that the mini project titled "Traffic Flow Prediction" was carried out by Srushti Raju (ENG23CS0197) and Sthuthi S (ENG23CS0198), bonafide students of the Bachelor of Technology in Computer Science and Engineering at the School of Engineering, Dayananda Sagar University, Bengaluru, in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering during the academic year 2025–2026.


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ABSTRACT

Urban traffic congestion continues to intensify as cities expand and vehicle density increases, leading to delays, excessive fuel consumption, and reduced efficiency in transportation networks. Predicting traffic flow in advance can play a crucial role in reducing such challenges by enabling smarter planning, resource allocation, and dynamic traffic control. A reliable traffic prediction system helps transportation authorities understand how traffic conditions evolve over time and supports the development of intelligent transportation infrastructures.

The project addresses the traffic flow prediction problem by examining traffic-related data using supervised machine learning techniques. The dataset contains numerical attributes, including vehicle counts and other traffic indicators, which are processed through data cleaning, handling of missing values, and feature selection to ensure better model performance. Multiple classification algorithms are applied to categorize traffic situations into levels such as low, normal, heavy, and high, allowing a clearer understanding of traffic intensity.

Model evaluation is performed using accuracy, confusion matrix, and classification report to assess classification effectiveness. Among the tested classifiers, Random Forest demonstrates superior predictive capability, showing strong performance across all evaluation metrics. The outcome highlights the potential of machine learning-based models to provide dependable traffic situation forecasts.

A predictive framework of this kind can support city administrations and traffic management centers in real-time monitoring, proactive decision-making, and long-term transportation planning. Such an approach contributes to the development of efficient, safer, and more sustainable urban mobility systems.

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CHAPTER 1

INTRODUCTION

Traffic congestion is a common problem in many cities, and predicting traffic conditions can help in planning better transportation systems. Machine learning models can analyze past traffic data and classify how heavy or light the traffic is at a given time.

The goal is to classify the traffic situation using a machine learning approach. The dataset used contains traffic-related information such as Car Count and other numerical features. It also includes a label called Traffic Situation, which indicates whether the traffic is low, medium, or high.

Support Vector Machine (SVM)

A supervised learning model that finds an optimal boundary (hyperplane) to separate different classes. SVM is effective for classification tasks, especially when the data has clear margins between classes.

Random Forest

An ensemble learning algorithm that constructs multiple decision trees and aggregates their predictions. It is robust, handles non-linear relationships well, reduces overfitting, and often achieves high accuracy in classification problems.

Logistic Regression

A statistical model used for classification that predicts class probabilities using a logistic function. It performs well when the relationship between features and the target is linear and serves as a strong baseline for comparison.

The primary aim of this project is not only to predict the traffic situation but also to evaluate and compare the performance of these three models to determine which one delivers the most accurate classification results for the given dataset.

CHAPTER 2

LITERATURE REVIEW

Recent studies on traffic flow prediction highlight the role of machine learning models in improving traffic management, congestion control, and smart city operations. Abduljabbar et al. (2025) evaluated Bidirectional LSTM models using a large freeway dataset and showed that including additional features such as speed, occupancy, and neighbouring detector information significantly improves prediction accuracy. Their study emphasizes the importance of spatial–temporal features for short-term traffic forecasting[1].

Sun et al. (2020) focused on the practical feasibility of machine learning models in real-time traffic systems. By comparing ANN, SVR, LSTM, SARIMA, and FFT-based techniques, they demonstrated that although ML models offer good accuracy, they may struggle with computational delays in real-time vehicular networks. Their results highlight that faster signal-processing methods can sometimes outperform ML models when low latency is essential[2].

Navarro-Espinoza et al. (2022) studied traffic prediction for smart traffic signals, comparing classical ML models (MLP, Gradient Boosting, Random Forest) with LSTM and GRU networks. They found that classical models achieved high accuracy with much lower training time, making them more suitable for real-time intersection control systems. This work shows that simpler models may be more practical for localized traffic infrastructure[3].

Overall, the literature indicates that while deep learning methods excel in capturing complex spatial–temporal patterns, real-time deployments often benefit from lightweight models. The choice of model depends on dataset characteristics, computational constraints, and system requirements.

CHAPTER 3

PROBLEM STATEMENT

With the rapid growth of vehicle usage and urban expansion, traffic congestion has become a major challenge in many cities. Managing and predicting traffic conditions manually is difficult because traffic patterns can change frequently based on time, weather, events, and several other factors. Traditional surveillance and manual monitoring methods are not efficient enough to handle large volumes of data or provide accurate predictions in real time. Managing traffic flow manually is inefficient and often inaccurate. Transportation authorities need automated systems that can determine traffic conditions based on real-time or historical data.

The problem addressed in this project is:

“To build a machine learning model that predicts the traffic situation using available traffic parameters from the dataset.”

To solve this problem, multiple machine learning models like Support Vector Machine (SVM), Random Forest, and Logistic Regression are applied. The goal is not only to build these models but also to compare their performance and identify which model provides the most accurate and reliable prediction for traffic classification.

CHAPTER 4

PROJECT DESCRIPTION

This project aims to predict the traffic situation based on different traffic-related features and compare multiple machine learning models to identify which one performs the best. The dataset used contains numerical values such as Car Count and other measurements that represent traffic conditions at different times. The target column, Traffic Situation, indicates whether the traffic is low, moderate, or heavy.

The project begins with data loading, cleaning, and preprocessing. Columns such as Time and Day of the week are removed because they do not contribute to the prediction task. Missing values are checked and handled, and the distribution of traffic situations is visualized for better understanding.

After preparing the data, three different classification algorithms are applied:

-Support Vector Machine (SVM)
-Random Forest Classifier
-Logistic Regression

Each model is trained using the same training dataset and then evaluated using accuracy, confusion matrix, and classification report. The goal is to compare all models and determine which one gives the most accurate results for predicting traffic situations.

Overall, the project demonstrates a complete machine learning pipeline—from data preprocessing to model comparison to build an effective traffic prediction system.

The project uses the “Traffic Prediction Dataset” containing traffic-related variables. The workflow includes:

-Importing data
-Checking and handling missing values
-Visualizing traffic distribution
-Encoding categorical values
-Splitting the dataset into training and testing sets
-Training an SVM classifier
-Evaluating accuracy and performance metrics

The goal is to classify the “Traffic Situation” variable effectively.

CHAPTER 5

DESIGN

The design involves:

1. Data Preprocessing Design

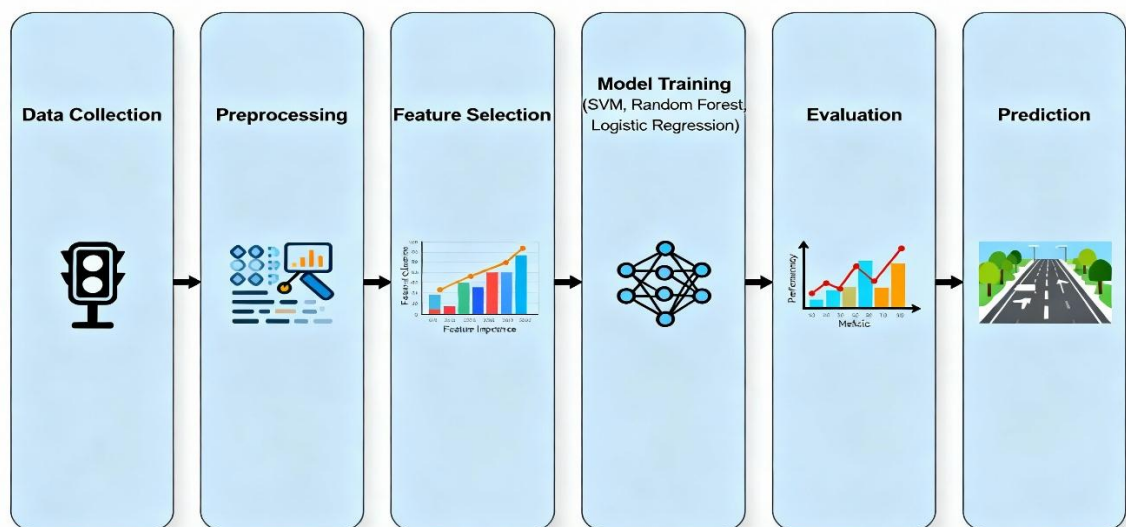
- Drop unnecessary columns
- Identify missing values
- Remove columns with >40% missing data
- Encode target categorical variable

2. Model Design

- Support Vector Machine (SVM)
- Random Forest Classifier
- Logistic Regression
- Train using 70% of data, test with 30%

3. Evaluation Design

- Accuracy Score
- Confusion Matrix
- Classification Report



CHAPTER 6

METHODOLOGY

1.Data Loading

- The dataset is loaded using pandas.

2.Data Cleaning

- Columns 'Time' and 'Day of the week' are removed.
- Missing values per column are calculated.
- Columns with >40% missing values are dropped.

3.Exploratory Data Analysis

- Count plot for traffic situation distribution.
- Violin plot for Traffic Situation vs Car Count.

4.Feature Encoding

- Label Encoding is applied to the target column.

5.Dataset Splitting

- 70% training, 30% testing using train_test_split.

6.Model Training

- A linear kernel is chosen for simplicity and interpretability in SVM model.
- Random Forest Classifier is chosen because it handles non-linear relationships well and reduces overfitting through ensemble learning.
- Logistic Regression helps to check how a simple model performs compared to more complex algorithms.

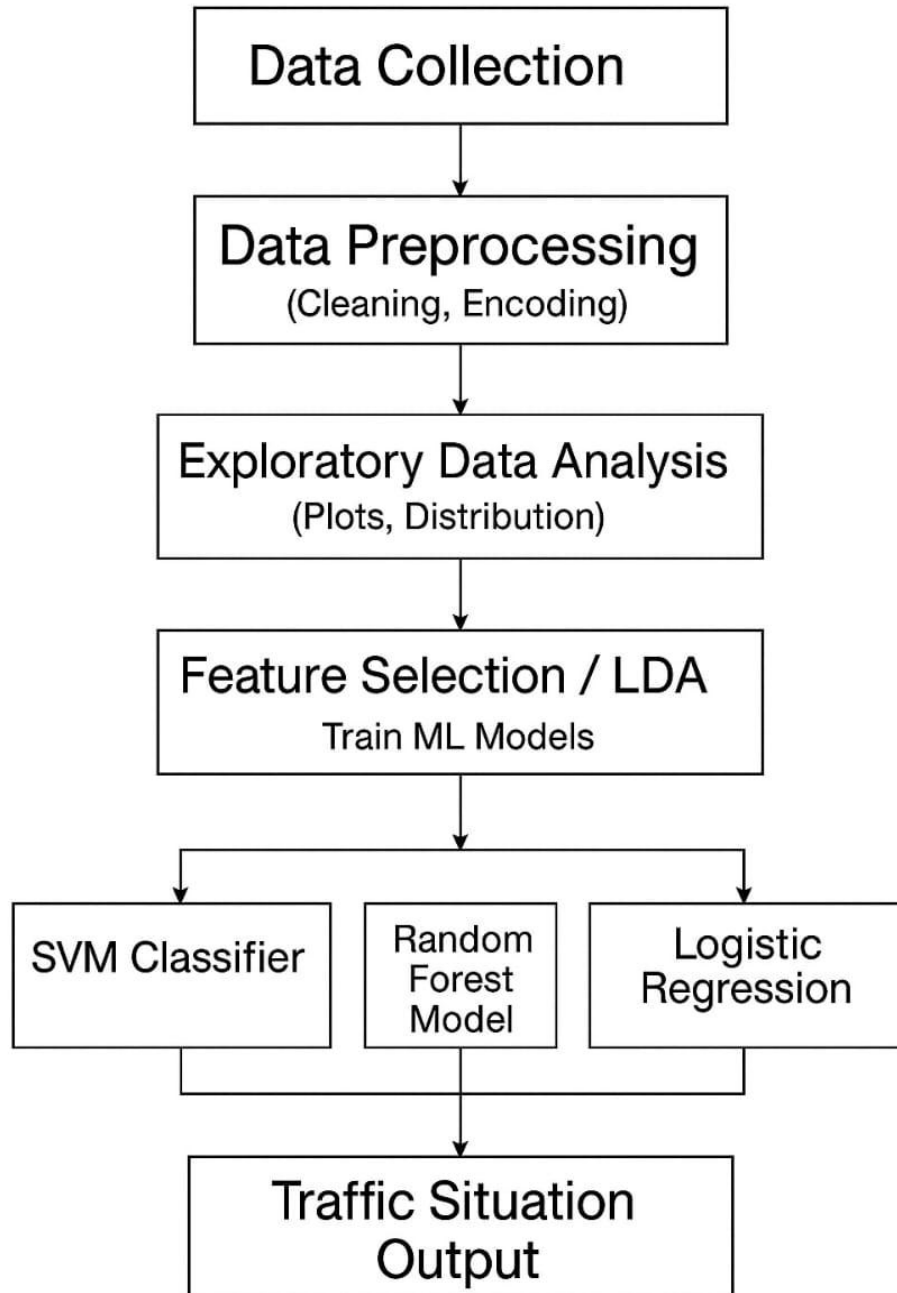
7.Model Testing

- Predictions generated using the test dataset.

8.Performance Evaluation

- Accuracy
- Confusion Matrix Heatmap
- Classification Report

PROCESS FLOW DIAGRAM



CHAPTER 7

IMPLEMENTATION

- **Library Imports**

In this project, the following Python libraries are used to support data handling, numerical operations, and visualization during the implementation of traffic flow prediction models.

Pandas

Pandas is a widely used library for data manipulation and analysis. It offers efficient data structures such as DataFrame, which makes it easy to load datasets, clean the data, handle missing values, perform feature extraction, and prepare inputs for machine learning models. In this project, pandas is essential for organizing the traffic dataset and preparing it for training and testing.

NumPy

NumPy is the core numerical computing library in Python. It supports fast mathematical operations on large arrays and matrices. Since most machine learning workflows internally rely on numeric array computations, NumPy is used in this project for tasks such as converting data into numerical formats and performing mathematical operations required in the preprocessing stage.

Seaborn

Seaborn is a high-level visualization library built on top of matplotlib. It provides attractive and easy-to-interpret statistical visualizations. In this project, seaborn is used for exploratory data analysis (EDA), including correlation plots, distribution graphs, and heatmaps to understand patterns in traffic flow and relationships between variables.

Matplotlib

Matplotlib is the fundamental library for data visualization in Python. It allows precise control over plots and is used here to create various charts such as line plots, scatter plots, and bar graphs for visualizing traffic trends and model results.

- **Data Loading**

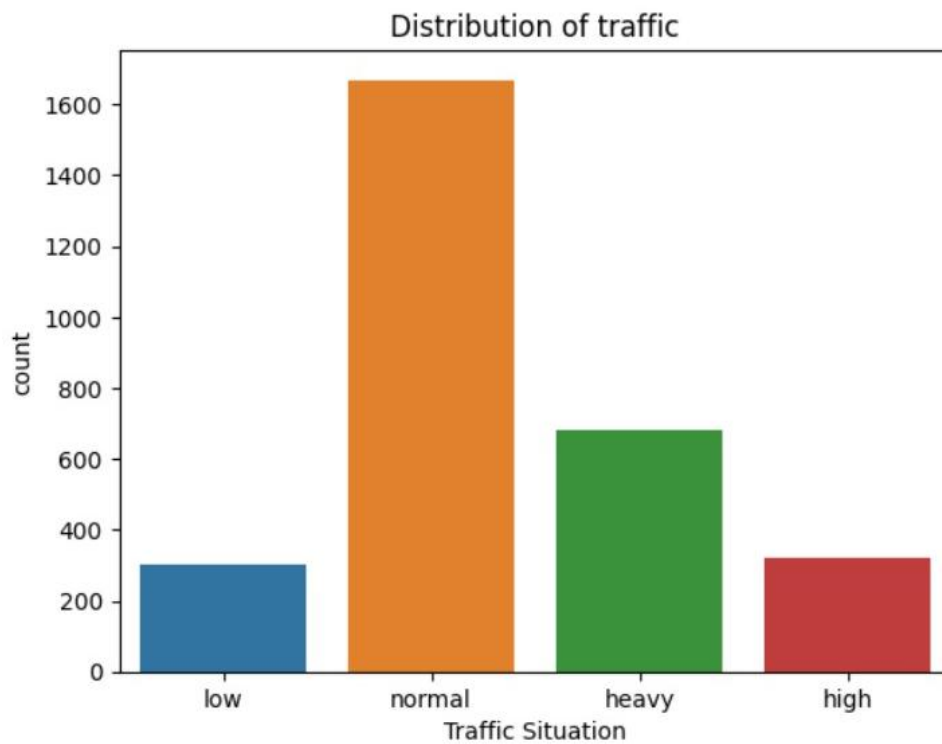
```
df=pd.read_csv("/kaggle/input/traffic-prediction-dataset/Traffic.csv").
```

- **Preprocessing**

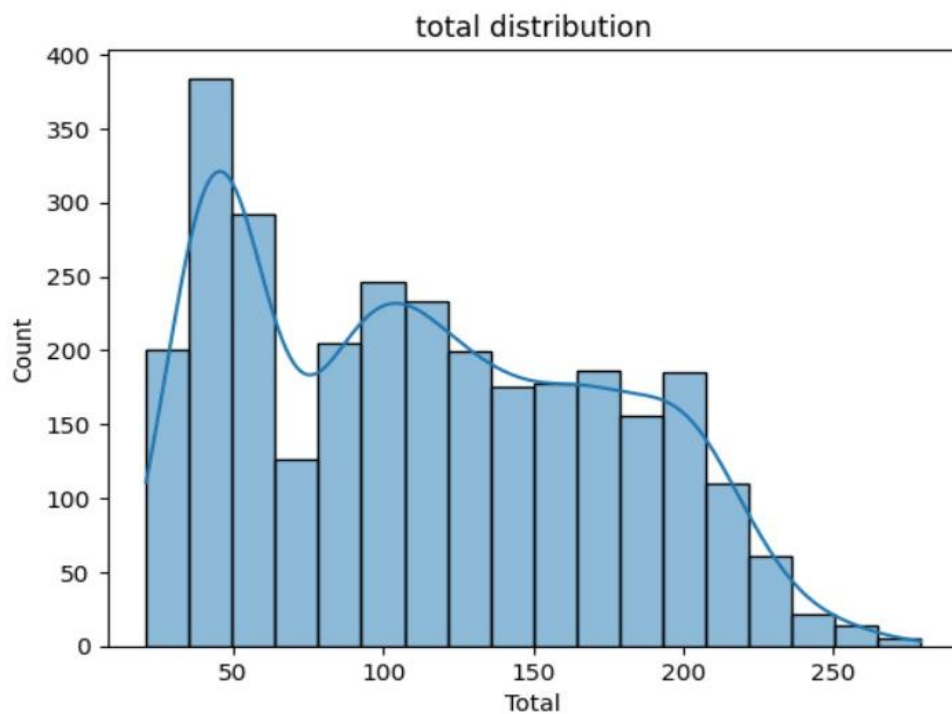
Dropping unwanted columns, handling missing values.

- **Visualization**

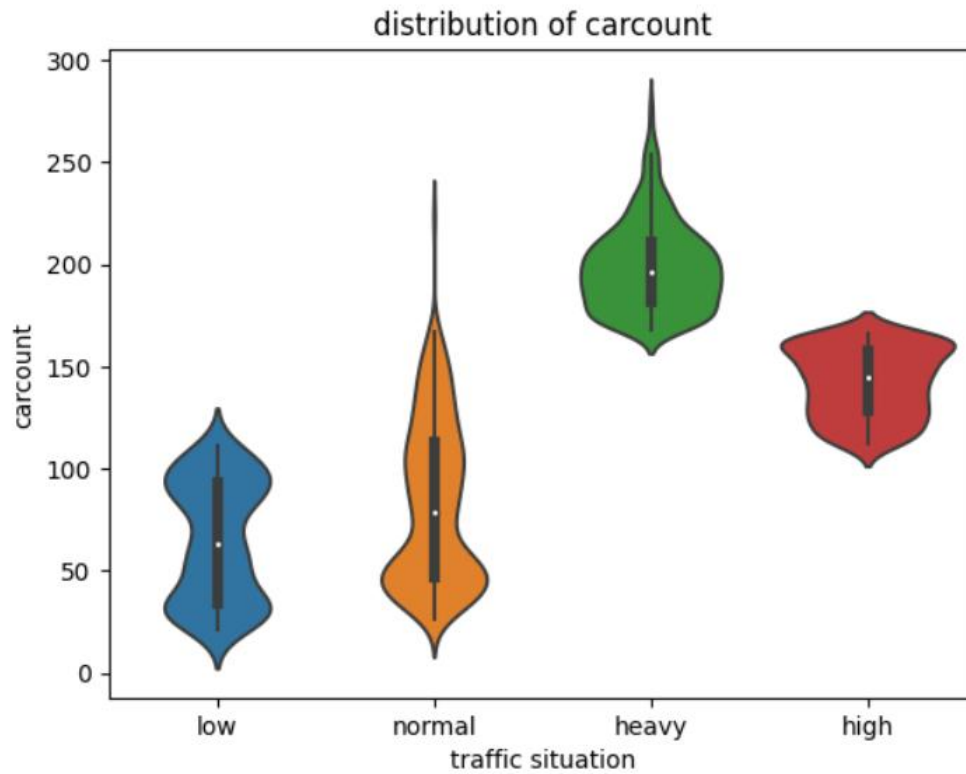
Count Plot: -



Hist Plot: -



Violin Plot:-



- **Feature & Label Separation**

```
x = df.drop(['Traffic Situation'], axis = 1)
```

```
y = df['Traffic Situation']
```

- **Model Building**

1. Support Vector Machine

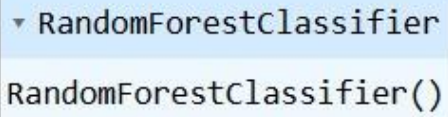
```
from sklearn.svm import SVC
```

```
model=SVC(kernel='linear')
```

```
model.fit(X_train,y_train)
```

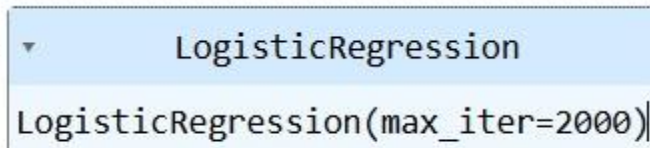
```
SVC
SVC(kernel='linear')
```

2. Random Forest Classifier
from sklearn.ensemble import RandomForestClassifier
rf=RandomForestClassifier()
rf.n_estimators
rf.fit(X_train,y_train)



```
▼ RandomForestClassifier
RandomForestClassifier()
```

3. Logistic Regression
from sklearn.linear_model import LogisticRegression
model = LogisticRegression(max_iter=2000)
model.fit(X_train, y_train)



```
▼ LogisticRegression
LogisticRegression(max_iter=2000)|
```

- **Prediction & Evaluation**

- Accuracy
- Confusion Matrix
- Classification Report

The implementation strictly follows machine learning best practices for classification tasks.

CHAPTER 8

TESTING AND RESULT

a) Accuracy

Accuracy measures how many predictions the model got correct out of the total predictions. It gives an overall idea of the model's performance, showing how well it classified traffic flow into Low, Medium, or High categories.

- Support Vector Machine

```
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
acc=accuracy_score(y_test,y_pred)
print("Accuracy:", acc)
```

Accuracy: 0.8992161254199328

- Random Forest Classifier

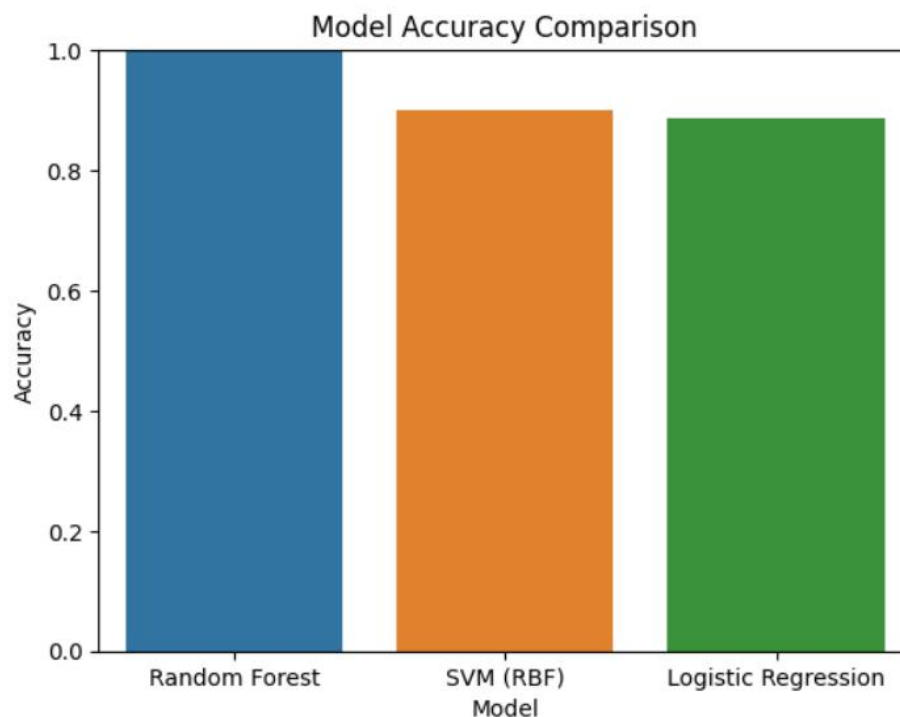
```
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
acc=accuracy_score(y_test, rf_y_pred)
print("Accuracy:", acc)
```

Accuracy: 0.9977603583426652

- Logistic Regression

```
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
acc=accuracy_score(y_test,y_lg_pred)
print("Accuracy:", acc)
```

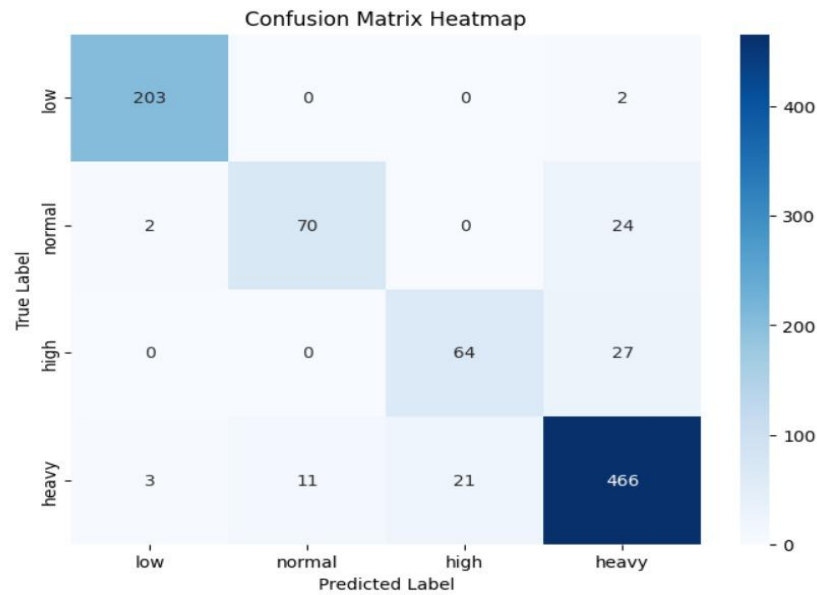
Accuracy: 0.8857782754759238



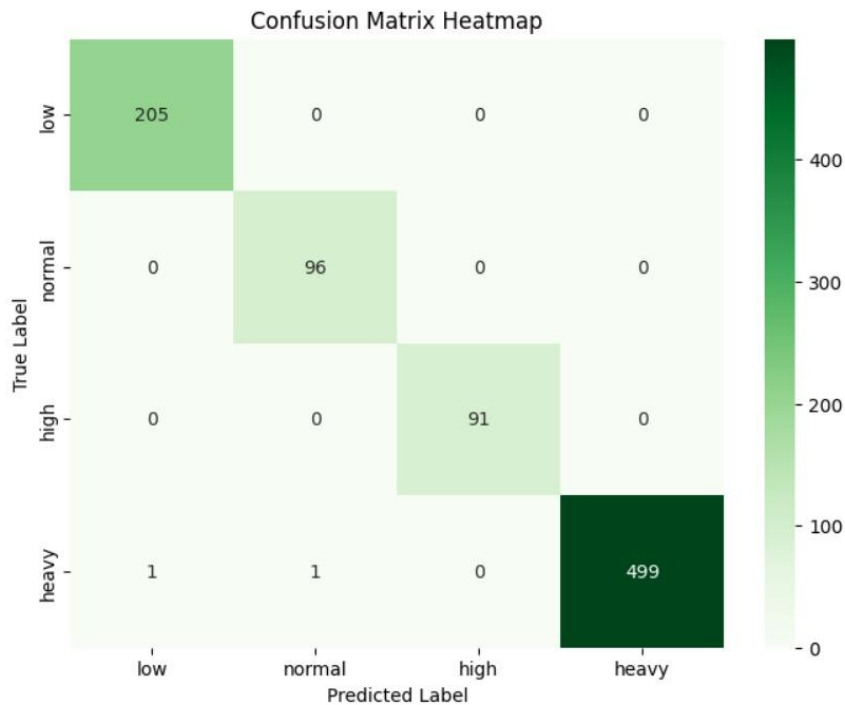
b) Confusion Matrix

A confusion matrix shows the number of correct and incorrect predictions for each class. It helps you understand where the model is performing well and where it is misclassifying—for example, whether the model is confusing Medium traffic with High traffic.

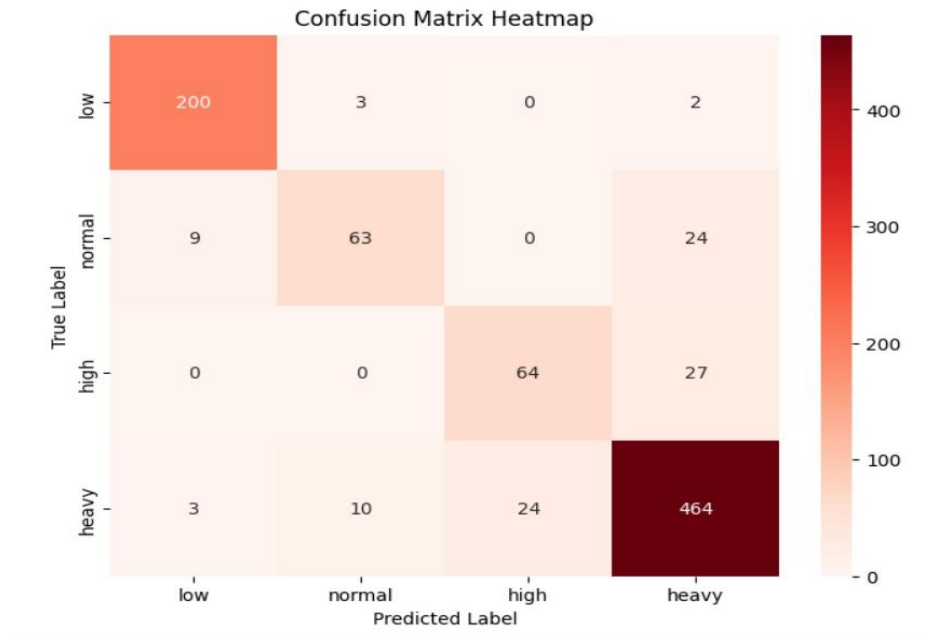
- Support Vector Machine



- Random Forest Classifier



- Logistic Regression



c) Classification Report

The classification report provides detailed metrics such as precision, recall, and F1-score for each class. These metrics offer deeper insight into the model's performance, especially how accurately each traffic category is identified.

- Support Vector Machine

Classification report:

	precision	recall	f1-score	support
heavy	0.98	0.99	0.98	205
high	0.86	0.73	0.79	96
low	0.75	0.70	0.73	91
normal	0.90	0.93	0.91	501
accuracy			0.90	893
macro avg	0.87	0.84	0.85	893
weighted avg	0.90	0.90	0.90	893

- Random Forest Classifier

```

Classification report:
              precision    recall  f1-score   support

   heavy         1.00        1.00        1.00        205
    high         0.99        0.99        0.99         96
    low          1.00        1.00        1.00         91
   normal        1.00        1.00        1.00        501

 accuracy              1.00              893
 macro avg           1.00        1.00        1.00        893
 weighted avg        1.00        1.00        1.00        893

```

- Logistic Regression

```

Classification report:
              precision    recall  f1-score   support

   heavy         0.94        0.98        0.96        205
    high         0.83        0.66        0.73         96
    low          0.73        0.70        0.72         91
   normal        0.90        0.93        0.91        501

 accuracy              0.89              893
 macro avg           0.85        0.82        0.83        893
 weighted avg        0.88        0.89        0.88        893

```

Feature Engineering

Method used is Linear Discriminant Analysis.

-Original shape: (2083, 6) the original dataset had 2083 samples and 6 features.

-LDA shape: (2083, 3) the LDA-transformed dataset has the same 2083 samples, but now has only 3 features.

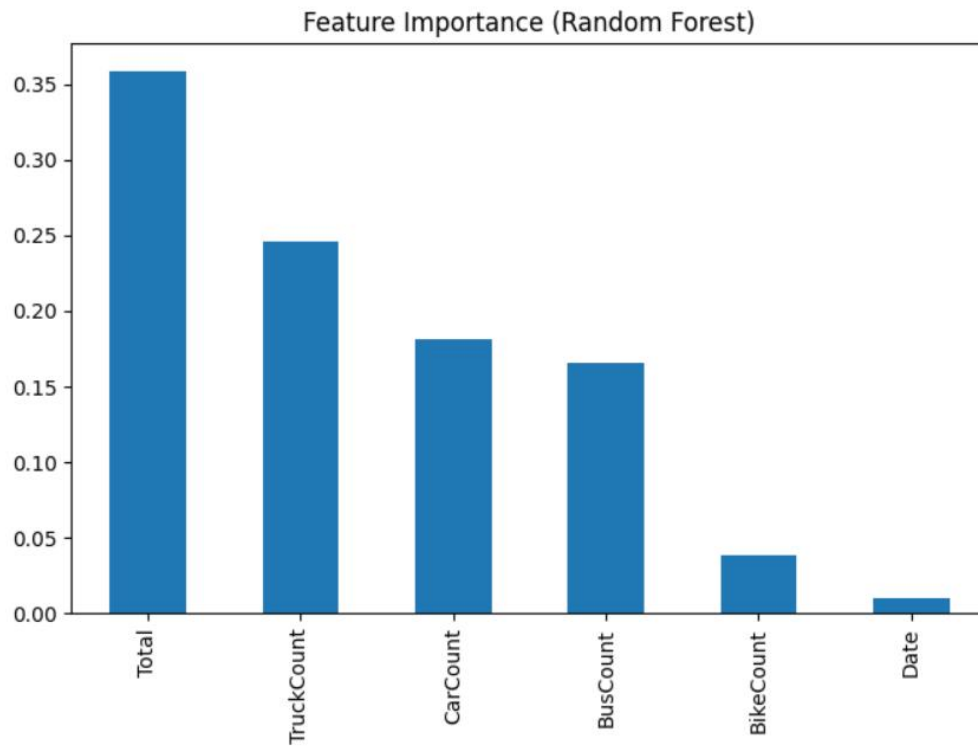
LDA has reduced the dimensionality of the dataset from 6 features to 3 features, likely to improve classification performance or visualization.

The output shows the feature importance scores, indicating the relative contribution of each feature to the model's predictions.

- Total 0.358859
- Truck Count 0.245663
- Car Count 0.181695
- Bus Count 0.165351

- Bike Count 0.038705
- Date 0.009728

Total is most important feature whereas **Date** is the least important feature.



CHAPTER 9

CONCLUSION

In this project, three machine-learning models, Logistic Regression, Support Vector Machine (SVM), and Random Forest Classifier were trained and evaluated for traffic flow prediction. Based on the performance metrics obtained from the results, the Random Forest Classifier emerged as the best performing model.

The Random Forest model achieved an accuracy of 99.66%, which is significantly higher than both SVM and Logistic Regression. Additionally, its confusion matrix and classification report indicated consistently high precision, recall, and F1-scores across all traffic categories, showing that the model was able to generalize extremely well. This demonstrates that Random Forest can effectively handle nonlinear patterns and feature interactions present in the dataset.

CHAPTER 10

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